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# A Control Matrix for Scientific Variable-Role Probing

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## Abstract

1 Interpretability requires experiments that distinguish a model mechanism from  
2 properties of the measurement. We examine this problem in linear probes for five  
3 variable roles in code models. The study includes three models and collectively  
4 covers all seven programming languages in XLCoST, but not every model-language  
5 combination. High probe accuracy supports different conclusions under matched  
6 controls. Misleading names reduce index probe F1 by 0.272 yet increase accumu-  
7 lator F1 by 0.079. Iterator probes can be refit after the same intervention in two  
8 models. On Python, boolean occurrence type is decodable at 0.981–0.988 macro-  
9 F1, but a masked source-line classifier reaches 0.983 without a language model.  
10 Rerunning both predictors with aligned predictions gives paired problem-clustered  
11 deltas of  $-0.003$  to  $+0.002$  across the three models, with intervals excluding  
12 any probe advantage above 0.014–0.017. This provides a bounded negative re-  
13 sult. Class or structure labels remain decodable after perturbation-specific refitting  
14 and pass a within-program label-permutation control in three models, yet a gate-  
15 specified activation patch at the query-name site fails in Qwen2.5-1.5B. These  
16 outcomes motivate a five-part control matrix. Decodability, probe-capacity control,  
17 cue sensitivity, surface sufficiency, and site-specific causal relevance are distinct  
18 claims, and clearing one does not imply clearing another. Reporting which com-  
19 parisons were run, what each found, and which remain untested makes a probing  
20 claim falsifiable.

## 21 1 Introduction

22 Linear probes are often used to ask what information a representation contains. A successful  
23 probe establishes decodability under a particular readout. It does not by itself establish that the  
24 model computed an abstract feature, that the feature is unavailable in surface form, or that the  
25 model uses it behaviorally [Belinkov, 2022, Voita and Titov, 2020]. Control tasks constrain probe  
26 capacity [Hewitt and Liang, 2019], and targeted studies show that probe accuracy need not entail task  
27 relevance [Ravichander et al., 2021, Elazar et al., 2021]. Capacity is only one source of ambiguity.  
28 Input leakage, label construction, sample resolution, and an unmatched intervention can each preserve  
29 high accuracy while changing the scientific claim.

30 This distinction is central to interpretability as a science. A measurement is useful when its target  
31 is explicit, its alternative explanations are tested, and contrary outcomes can change the conclusion.  
32 We use variable-role probing as a comparative case study. Variable roles describe how an identifier  
33 functions in a program, such as accumulating values or indexing a container [Sajaniemi, 2002].  
34 They are attractive probe targets because the same nominal role can be expressed by names, syntax,  
35 and distributed context. That multiplicity also makes probe accuracy underidentified. Prior code-  
36 probing studies, including the 15-task INSPECT framework, map decodable syntactic and semantic  
37 information across layers and models [Karmakar and Robbes, 2021, Troshin and Chirkova, 2022,  
38 Karmakar and Robbes, 2024]. We do not propose another taxonomy. We contribute a role-by-control

Table 1: Model and language coverage. No row should be read as a complete three-model by seven-language experiment. Table 3 gives the comparison coverage for each role.

| Role               | Models                               | Language coverage                         |
|--------------------|--------------------------------------|-------------------------------------------|
| Accumulator        | Qwen2.5-1.5B                         | Python plus Java, C++, C, C#              |
| Index or key       | Qwen2.5-1.5B                         | Python, Java, JavaScript, C++, C, C#      |
| Iterator           | Qwen2.5-Coder-1.5B,<br>StarCoder2-7B | Python, Java, JavaScript, PHP, C++, C, C# |
| Boolean            | All three                            | Python, Java, JavaScript, PHP             |
| Class or structure | All three                            | Python, C++, JavaScript, C                |

evidence map that records which interpretation survives, which comparison is incommensurate, and what outcome would falsify each surviving claim.

Our evidence covers five roles, three models, and the seven-language XLCoST corpus [Zhu et al., 2022]. The experiment is intentionally reported as an uneven evidence matrix rather than a full factorial study. Qwen2.5-1.5B, Qwen2.5-Coder-1.5B, and StarCoder2-7B all appear in the controlled boolean and class or structure experiments [Qwen et al., 2024, Hui et al., 2024, Lozhkov et al., 2024]. Earlier accumulator and index runs use one model and a weaker split protocol. Iterator runs cover Qwen2.5-Coder-1.5B and StarCoder2-7B across all seven languages. This is a methodological case study of evidential boundaries, not a validation of the matrix as a universal framework. Its heterogeneity separates replicated findings from leads that still require confirmation.

The central result is not that one representation type wins. The same headline observation, high linear-probe F1, produces different evidential outcomes. Index probes depend strongly on names. Accumulator and iterator probes can be refit after misleading renames. A model-free classifier matches boolean probe scores, and paired intervals exclude a probe advantage above 0.017. Class or structure probe performance remains stable under tested renames and passes a randomized-label control, but the gate-specified causal intervention fails at one site. A single accuracy number cannot identify among these explanations.

## 2 Study design and evidence matrix

**Roles, models, and languages.** We treat five roles as related case studies, not measurement-equivalent instances of one construct. Accumulators update within loops. Indices or keys appear as loop counters or subscripts. Iterators are loop-bound names and their uses. Boolean labels distinguish assignment, conditional, loop, and return uses. Class or structure labels mark declared type names and references. Binary token probes label all other tokens as negative. Heuristic AST and regular-expression extractors vary by language and lack a common multilingual gold set. Protocols also differ. Accumulator, index, and iterator use legacy token-stratified splits and test-maximized layers. Boolean mean-pools occurrences, uses problem-grouped splits, and predicts usage type. The class or structure pipeline uses token labels, problem groups, and five seeds. The modern pipelines select layers on validation data. XLCoST contains C, C++, C#, Java, JavaScript, PHP, and Python. Table 1 states actual coverage. Scores across rows do not estimate one common task.

**Four control questions.** Renaming asks whether lexical identity carries the prediction. We compare original names with numeric, single-character, collapsed, random-noun, and misleading names drawn from another role. Each condition refits its probe and selects its layer, so preserved F1 means that a similarly predictive readout can be recovered after renaming. It does not show that one fixed feature direction is invariant. A model-free character classifier with the target name masked asks whether local source text already determines the label. Randomized-label controls ask whether the probe family can fit a specified null task. The boolean null assigns a random label per variable name, whereas the class or structure null permutes labels within programs. Activation patching asks whether replacing a residual state at a gate-specified site moves a behavioral logit readout [Meng et al., 2022].

These controls are not interchangeable. Renaming can show name robustness while leaving syntax untouched. With refitting, it measures recoverability rather than invariance. A positive selectivity score can reject its specified null task while leaving a surface shortcut intact. A model-free com-

Table 2: Probe results after the strongest available control. Values are macro-F1. Deltas compare the misleading-name condition with original names.

| Role               | Baseline probe | Key control result                   | Strongest supported inference                                 |
|--------------------|----------------|--------------------------------------|---------------------------------------------------------------|
| Index              | 0.959          | $\Delta = -0.272$                    | Refit performance depends on names in the legacy Qwen run     |
| Accumulator        | 0.893          | $\Delta = +0.079$                    | Refit performance survives that rename in the legacy run      |
| Iterator           | 0.988, 0.990   | $\Delta = +0.012, +0.009$            | Refit performance survives misleading names in two models     |
| Boolean            | 0.981–0.988    | paired $\Delta \in [-0.003, +0.002]$ | Probe advantage over masked local source excluded above 0.017 |
| Class or structure | 0.979–0.982    | $ \Delta  \leq 0.004$                | Refit robustness plus within-program permutation selectivity  |

80 parator can establish surface sufficiency but cannot show how the model uses the feature. A causal  
81 intervention can ground a use claim only at its specified layer, span, direction, and readout.

82 **Evidence grades.** Boolean and class or structure experiments use problem-grouped splits,  
83 validation-selected layers, five seeds, and paired or clustered uncertainty. Iterator artifacts use  
84 two models but retain the legacy token-level split and test-maximized layer, without matched surface  
85 or capacity controls. Accumulator and index likewise use token-level splits, test-maximized layers,  
86 and whole-word renaming. We retain their large, opposite effects as a falsifiable hypothesis, not as a  
87 confirmed estimate of a population difference. A DeepSeek index run is excluded because a tokenizer  
88 offset error invalidates its activation alignment.

### 89 3 The same probe score supports different claims

#### 90 3.1 Lexical intervention separates roles

91 Perturbation-specific refitting produces opposite effects for index and accumulator probes. The  
92 Qwen2.5-1.5B index probe falls from 0.959 to 0.687, a change of  $-0.272$ . The accumulator probe  
93 rises from 0.893 to 0.972, a change of  $+0.079$ . The intervention therefore does more than lower  
94 overall data quality. It changes how recoverable the two labels are under the legacy readout. This is  
95 the study’s clearest dissociation, but occurrence-level splitting, test-maximized layers, and refitting  
96 prevent a fixed-direction invariance interpretation. This effect-size estimate remains exploratory.

97 The iterator experiments provide separate evidence of refit robustness at near-ceiling baseline accuracy.  
98 Qwen2.5-Coder-1.5B and StarCoder2-7B reach baseline F1 of 0.988 and 0.990. Misleading iterator  
99 names change those scores by  $+0.012$  and  $+0.009$ . Python-trained probes transfer across the other  
100 six XLCOST languages with F1 ranges of 0.936–0.983 and 0.906–0.964. These results support  
101 recoverability after the tested naming intervention. They do not establish feature invariance or  
102 abstraction because the probe and best layer are refit and the iterator runs lack a matched model-free  
103 surface comparator.

#### 104 3.2 A model-free baseline bounds the boolean result

105 Boolean occurrence type illustrates why a high, replicated probe score can still provide limited  
106 scientific evidence. On a frozen Python sample, the three models reach 0.982, 0.981, and 0.988  
107 macro-F1. Selectivity ranges from  $+0.272$  to  $+0.288$ . These values reject the specific hypothesis that  
108 an equally expressive probe predicts labels randomly assigned per variable name as well as it predicts  
109 the real usage labels. Yet a character classifier on the enclosing source line, with the variable name  
110 masked, reaches 0.983. The probe-minus-surface differences are  $-0.001$ ,  $-0.002$ , and  $+0.005$ .

111 The original exports retained no aligned predictions, so we reran both predictors on the frozen sample  
112 and retained aligned predictions. Before pairing, the reruns reproduced the committed scores. The  
113 paired probe-minus-surface estimates, followed by their problem-clustered 95% intervals, are  $-0.002$   
114  $[-0.017, 0.014]$ ,  $-0.003 [-0.037, 0.014]$ , and  $+0.002 [-0.016, 0.017]$  over 915 problem clusters.  
115 Every interval covers zero, and each excludes a probe advantage larger than 0.014–0.017 macro-F1.

Five renaming conditions also change refit performance by at most 0.013. The supported claim is now sharper. Boolean occurrence type is decodable after the tested renames. The paired intervals exclude a probe advantage above 0.014–0.017 over masked local syntax on Python. The cross-language runs extend decodability to Java, JavaScript, and PHP. Because only Python has renaming runs, the cross-language results do not extend the refit-robustness claim beyond Python.

### 3.3 A causal null limits the class or structure claim

The class or structure experiments and the boolean experiments use different controls. Across all three models, baseline F1 is 0.979–0.982 and within-program permutation selectivity is +0.503–+0.553. Misleading class-style names change F1 by −0.004, −0.004, and +0.004. Python-trained probes also transfer to C++, JavaScript, and C at 0.885–0.980. This supports a permutation-selective signal whose predictive performance can be recovered after the tested renames. It does not establish a fixed rename-invariant direction.

We then test a stronger causal claim in Qwen2.5-1.5B. The experiment contains 288 matched class-versus-function prompts in 48 template clusters. Let  $D = \ell(\text{True}) - \ell(\text{False})$ . Denoising patches the class-source residual into a function prompt and defines  $e_d = D_{\text{patched function}} - D_{\text{function}}$ . Noising reverses the patch and defines  $e_n = D_{\text{class}} - D_{\text{patched class}}$ . Recovery is the ratio of the mean signed effect to the mean matched gap  $D_{\text{class}} - D_{\text{function}}$ . Percentile intervals resample template clusters. The gate additionally requires positive, control-separated effects, recovery of at least 0.05, and mean effects of at least 0.10 in both directions.

Both prompt classes prefer True, which limits the behavioral readout’s effective decision range. Their mean logit differences are +2.75 for class prompts and +1.78 for function prompts, although the matched class-minus-function gap is +0.97 and has the expected sign for 98.6% of pairs. The gate therefore tests whether patching transfers this relative distinction, not whether it flips a well-calibrated binary decision.

Denoising produces a mean effect of 0.0087 with a 95% interval [−0.0007, 0.0188] and recovery 0.0089. Noising produces 0.0199 [0.0109, 0.0296] with recovery 0.0204. The intervention therefore fails the specified gate. The protocol and results entered the repository together, so we do not claim auditable preregistration. This is a site-specific causal null, not evidence that class information is absent or unused everywhere. More directly, the intervals exclude mean effects above 0.0188 and 0.0296 logit-difference units in the two tested directions. Strong decodability, positive selectivity, and renaming robustness therefore do not establish a substantive bidirectional contribution at this site and readout.

## 4 A claim–control matrix for scientific probing

The experiments distinguish logically separate claims rather than a cumulative ladder.

1. **Decodability.** A held-out probe exceeds a simple label baseline.
2. **Probe-capacity control.** The probe exceeds a matched randomized-label task whose construction is reported. This implies neither training dependence nor surface irreducibility.
3. **Cue sensitivity.** A fixed probe on a paired rename isolates one question, whereas perturbation-specific refitting measures recoverability after the cue changes.
4. **Surface sufficiency.** A model-free comparator on the same population tests whether surface information already predicts the label, and a representational advantage further requires uncertainty on the paired difference.
5. **Site-specific causal relevance.** A gate-specified intervention changes a behavioral readout in both directions and separates from placebo and random controls.

No claim entails another. Table 3 records which comparisons were run for each role. Only boolean has a matched surface comparator, which finds masked local syntax sufficient. Only class or structure has a causal gate, which the intervention fails at the selected site.

**Falsifying outcomes.** The framework is useful only if results can change the interpretation. A problem-grouped rerun could shrink or reverse the accumulator-index dissociation. An all-layer class or structure patch could reveal a causal site outside the probe-selected layer. A seven-language

Table 3: Comparison coverage by role. Check marks denote modern comparisons, circles denote legacy comparisons awaiting replication, and NR denotes comparisons not run. Outcomes appear in the text and Table 4.

| Comparison               | Accum. | Index | Iterator | Boolean | Class |
|--------------------------|--------|-------|----------|---------|-------|
| Decodability             | ○      | ○     | ○        | ✓       | ✓     |
| Randomized-label control | NR     | NR    | NR       | ✓       | ✓     |
| Rename intervention      | ○      | ○     | ○        | ✓       | ✓     |
| Surface comparator       | NR     | NR    | NR       | ✓       | NR    |
| Causal gate              | NR     | NR    | NR       | NR      | ✓     |

Table 4: Scientific status of the paper’s main interpretations. “Open” means that the required comparison has not been run under a matched protocol.

| Interpretation                                                          | Evidence in this study                                                     | Status and decisive next test                                                 |
|-------------------------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Legacy roles may use different cues                                     | Opposite refit deltas in incommensurate pipelines, plus iterator stability | Exploratory. Repeat with problem groups, scope-aware renaming, and intervals  |
| Boolean exceeds Python local syntax                                     | Paired reruns, three models, 915 problem clusters                          | Not supported. 95% intervals exclude advantages above 0.014–0.017             |
| Class refit performance survives renaming                               | Three-model permutation selectivity and misleading-name deltas near zero   | Supported as recoverability. Evaluate one fixed probe across paired renames   |
| Query-name signal makes a substantive bidirectional causal contribution | Two-direction patching with placebo and random controls                    | Not supported at Qwen layer 18. Run an explicitly exploratory all-layer sweep |
| Iterator exceeds local surface form                                     | High F1 in two models with refit robustness, no model-free comparator      | Open. Run a matched masked-source comparator on that sample                   |
| Cross-language role abstraction                                         | Strong transfer with uneven role and model coverage                        | Open. Apply one extractor and control battery to a complete language grid     |

166 rename pipeline could reveal language-specific cue dependence. Each outcome would revise a named  
167 claim.

168 **Replication standard.** We recommend publishing the evidence matrix with the metric table. Every  
169 cell should identify the prediction unit, label, model, corpus, split, layer selection, seeds, comparator,  
170 uncertainty estimand, and intervention gate. Negative results should state the smallest effect the  
171 design can exclude.

## 172 5 Identifiability and evaluation design

173 Matched contrasts determine what each control identifies. The paired boolean intervals bound probe-  
174 minus-surface differences. Refit renaming measures recoverability. The class or structure patch fails  
175 the specified bidirectional gate at one site. None supports a broader invariance or causal claim.

## 176 6 Limitations

177 The study collectively covers three models, five roles, and seven languages, but not a complete  
178  $3 \times 5 \times 7$  design, and Table 1 gives the per-role model and language coverage.

179 Legacy accumulator-index results predate problem-grouped splits and scope-aware renaming, and  
180 function overlap, textual replacement, layer selection, and refitting may inflate the effect. Iterator  
181 has no matched surface or capacity comparator, and its legacy artifacts record only aggregate scores.  
182 Boolean labels are nearly predictable from local form, with paired 95% intervals whose upper  
183 endpoints do not exceed 0.017. Class or structure patching covers one model, layer, site, and readout,  
184 and stopping after the Qwen null was not registered in advance. Other probes remain correlational,  
185 and cross-language transfer may reflect shared syntax, names, or extractor artifacts.

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## 249 A Per-role control results

250 **Boolean probe figures.** The baseline panels retain the individual surface comparators hidden by  
 251 the best-baseline table. Renaming is available only for Python. The absence of corresponding figures  
 252 in other languages is a coverage hole.

Table 5: Complete boolean occurrence-type probe and model-free baseline results. The best baseline is selected from name-only, masked statement, masked line, masked window, covariate-only, and majority features.  $\rho$  is a descriptive one-instance score step, not an interval or equivalence test. Aligned probe/baseline predictions were not retained.

| language   | model              | problems | probe F1 | best baseline | difference | $\rho$ | selectivity |
|------------|--------------------|----------|----------|---------------|------------|--------|-------------|
| Java       | Qwen2.5-1.5B       | 1,427    | 0.982    | 0.980         | +0.002     | 0.0120 | +0.255      |
| Java       | Qwen2.5-Coder-1.5B | 1,427    | 0.976    | 0.980         | -0.004     | 0.0079 | +0.271      |
| Java       | StarCoder2-7B      | 1,427    | 0.990    | 0.980         | +0.010     | 0.0120 | +0.243      |
| JavaScript | Qwen2.5-1.5B       | 1,351    | 0.989    | 0.992         | -0.003     | 0.0170 | +0.132      |
| JavaScript | Qwen2.5-Coder-1.5B | 1,351    | 0.998    | 0.992         | +0.006     | 0.0170 | +0.150      |
| JavaScript | StarCoder2-7B      | 1,351    | 0.988    | 0.992         | -0.004     | 0.0170 | +0.120      |
| PHP        | Qwen2.5-1.5B       | 1,305    | 0.983    | 0.969         | +0.014     | 0.0080 | +0.214      |
| PHP        | Qwen2.5-Coder-1.5B | 1,305    | 0.975    | 0.969         | +0.006     | 0.0080 | +0.216      |
| PHP        | StarCoder2-7B      | 1,305    | 0.990    | 0.969         | +0.021     | 0.0080 | +0.240      |
| Python     | Qwen2.5-1.5B       | 1,301    | 0.982    | 0.983         | -0.001     | 0.0068 | +0.280      |
| Python     | Qwen2.5-Coder-1.5B | 1,301    | 0.981    | 0.983         | -0.002     | 0.0068 | +0.272      |
| Python     | StarCoder2-7B      | 1,301    | 0.988    | 0.983         | +0.005     | 0.0068 | +0.288      |

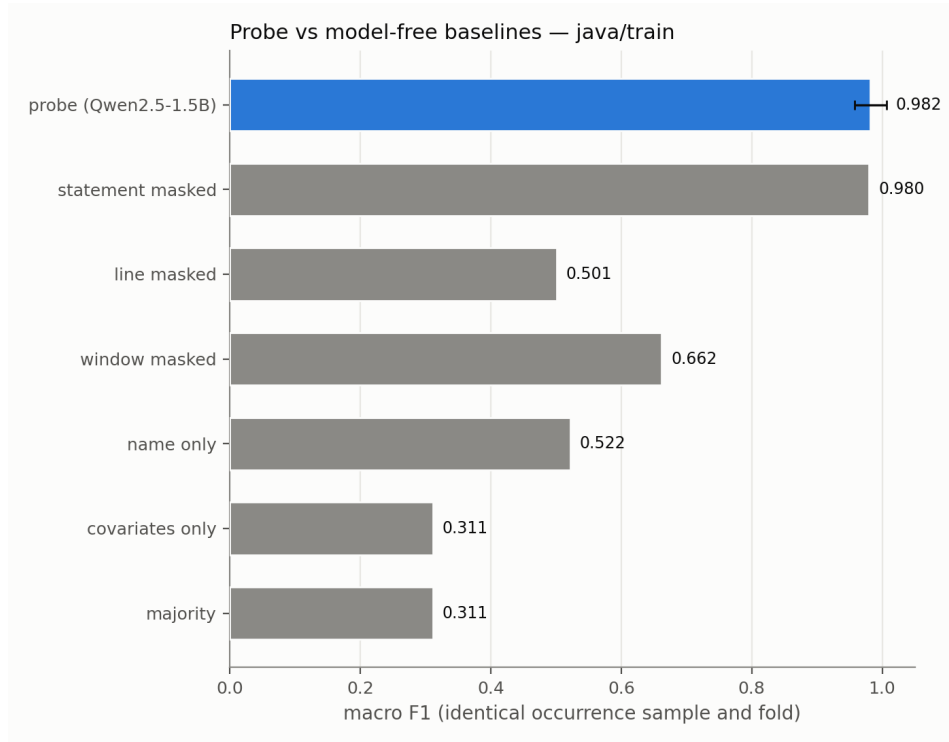


Figure 1: Boolean probe and model-free baselines for Java, Qwen2.5-1.5B.



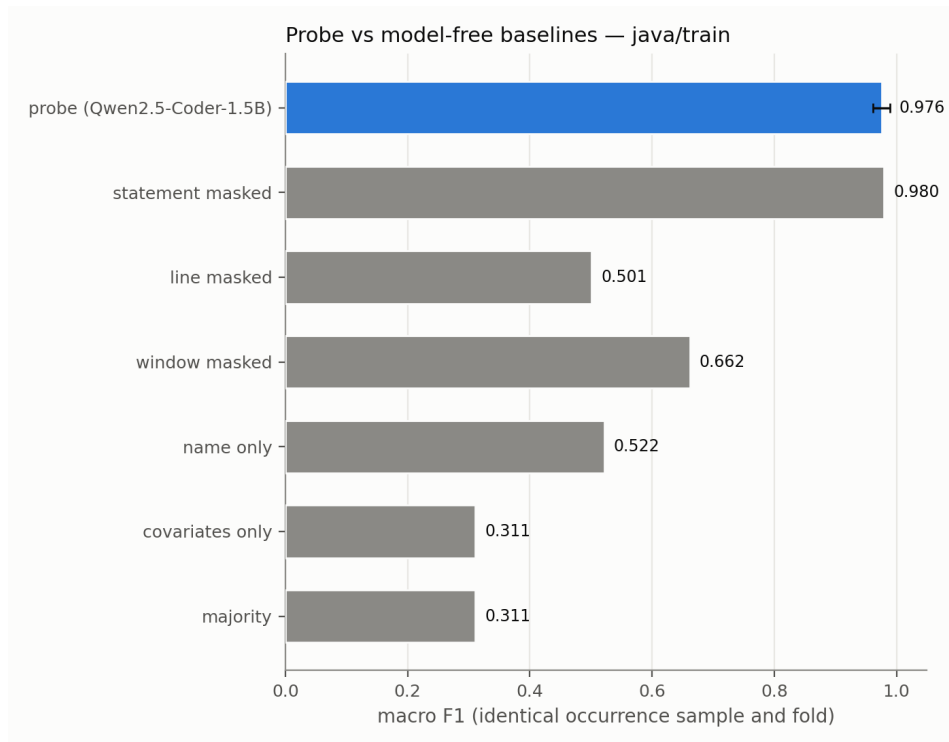


Figure 2: Boolean probe and model-free baselines for Java, Qwen2.5-Coder-1.5B.

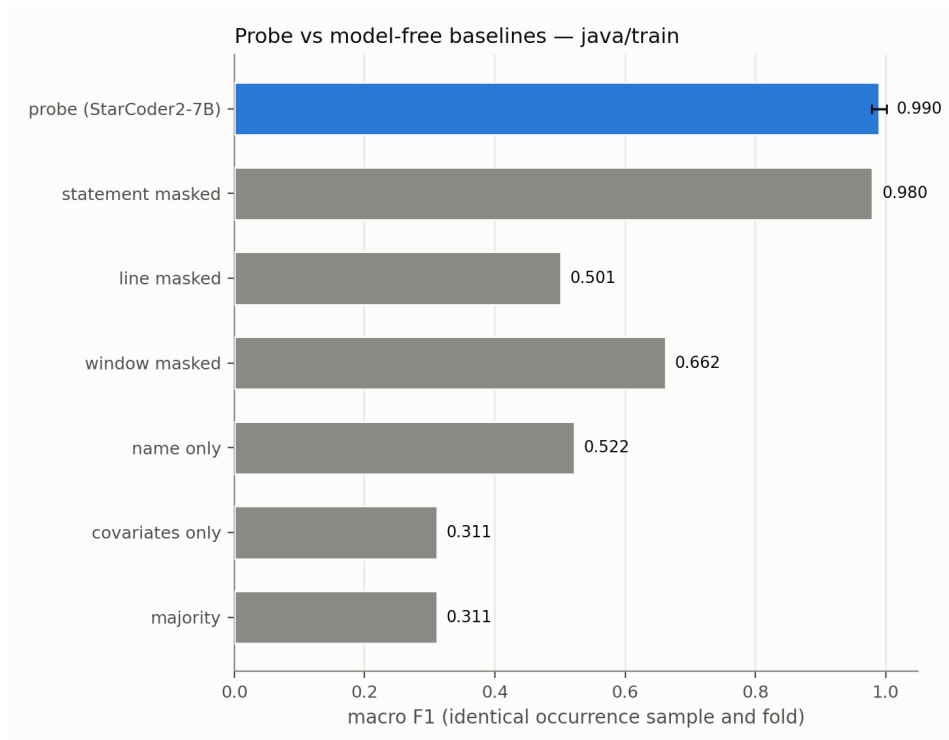


Figure 3: Boolean probe and model-free baselines for Java, StarCoder2-7B.

Table 6: Boolean Python identifier-renaming audit. C1–C5 are the five committed renaming conditions. Each condition refits a probe and reselects its layer, so deltas measure recoverability rather than fixed-direction invariance. Intervals are problem-clustered. Empty renaming fields for other languages indicate that those runs were not committed, not a null result.

| model              | C1     | C2     | C3     | C4     | C5     |
|--------------------|--------|--------|--------|--------|--------|
| Qwen2.5-1.5B       | +0.003 | -0.004 | +0.001 | -0.003 | +0.001 |
| Qwen2.5-Coder-1.5B | +0.001 | -0.010 | +0.006 | -0.004 | -0.006 |
| StarCoder2-7B      | -0.003 | -0.013 | -0.006 | -0.001 | -0.006 |

Table 7: Complete model-free boolean baselines by language. Baselines are shared across the three neural models because they use the same frozen examples and surface features.

| language   | majority | name  | statement | line  | window | covariates |
|------------|----------|-------|-----------|-------|--------|------------|
| Java       | 0.311    | 0.522 | 0.980     | 0.501 | 0.662  | 0.311      |
| JavaScript | 0.227    | 0.373 | 0.992     | 0.369 | 0.552  | 0.226      |
| PHP        | 0.228    | 0.505 | 0.969     | 0.413 | 0.581  | 0.228      |
| Python     | 0.217    | 0.422 | 0.970     | 0.983 | 0.605  | 0.216      |

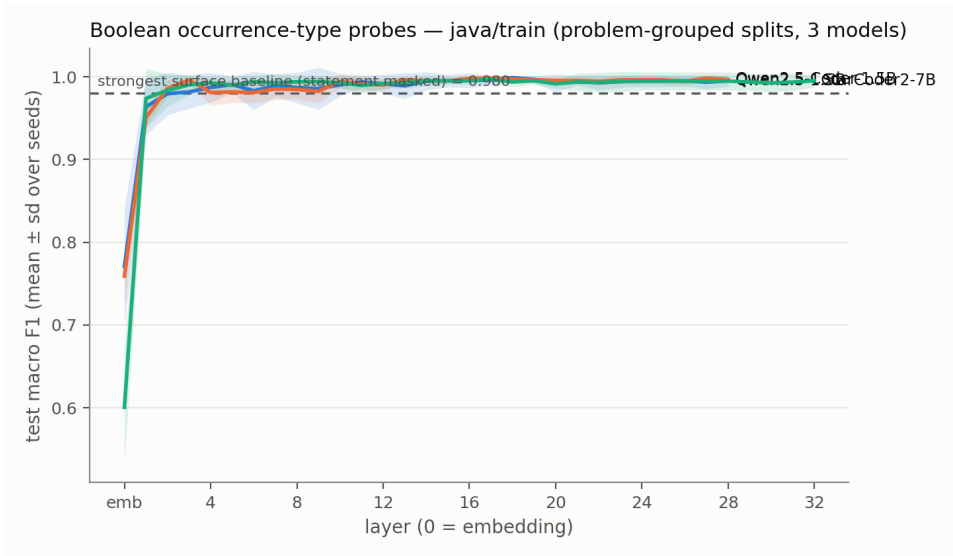


Figure 4: Boolean probe layer curves for Java across the three models.

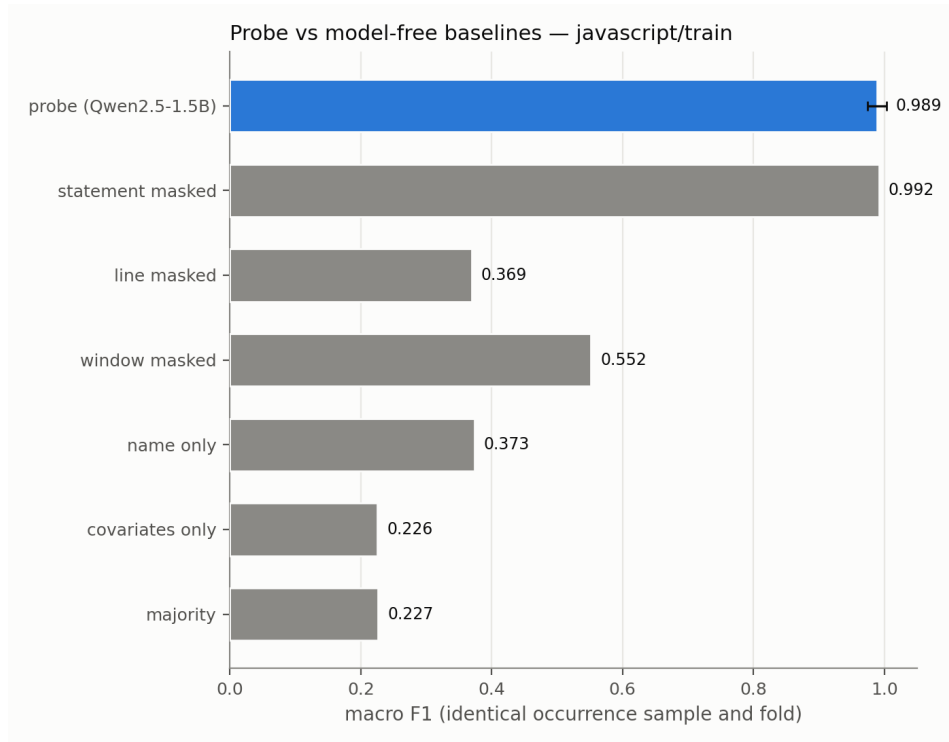


Figure 5: Boolean probe and model-free baselines for JavaScript, Qwen2.5-1.5B.

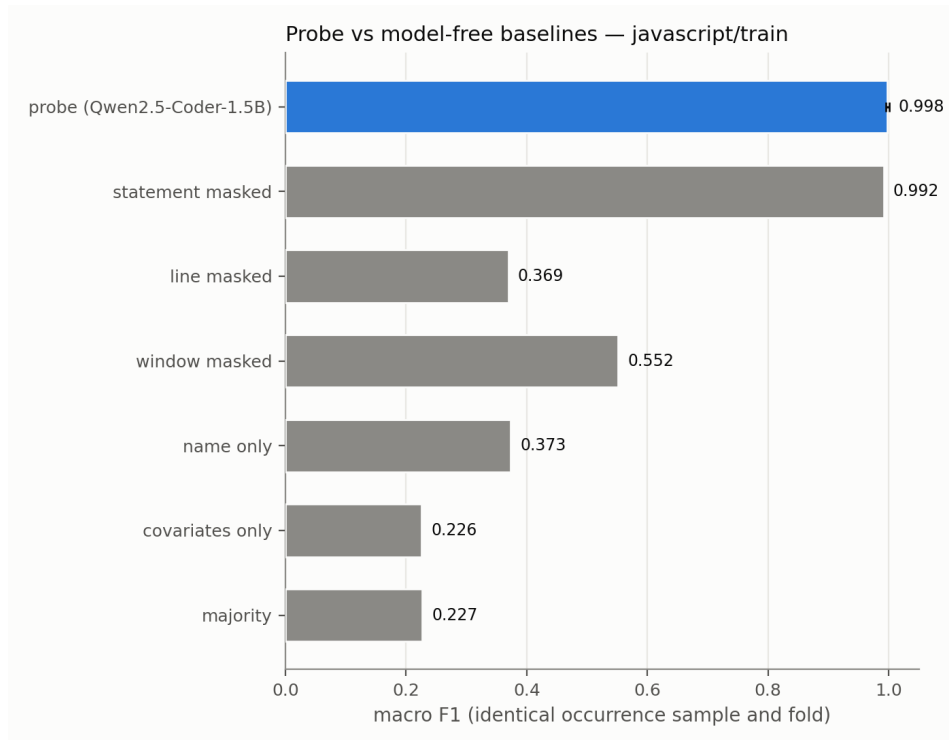


Figure 6: Boolean probe and model-free baselines for JavaScript, Qwen2.5-Coder-1.5B.

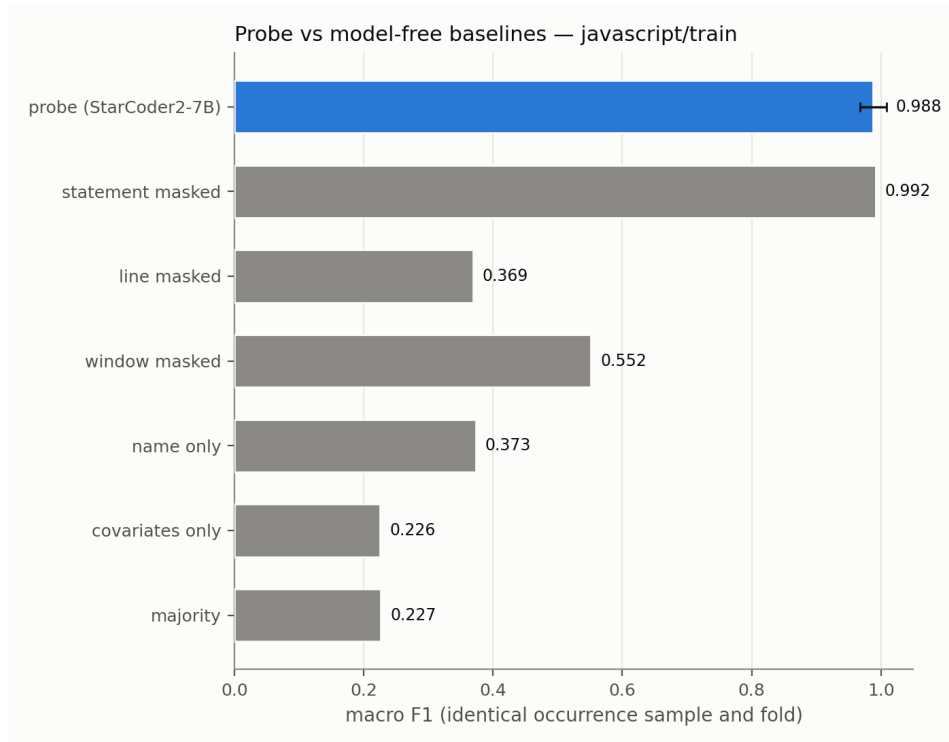


Figure 7: Boolean probe and model-free baselines for JavaScript, StarCoder2-7B.

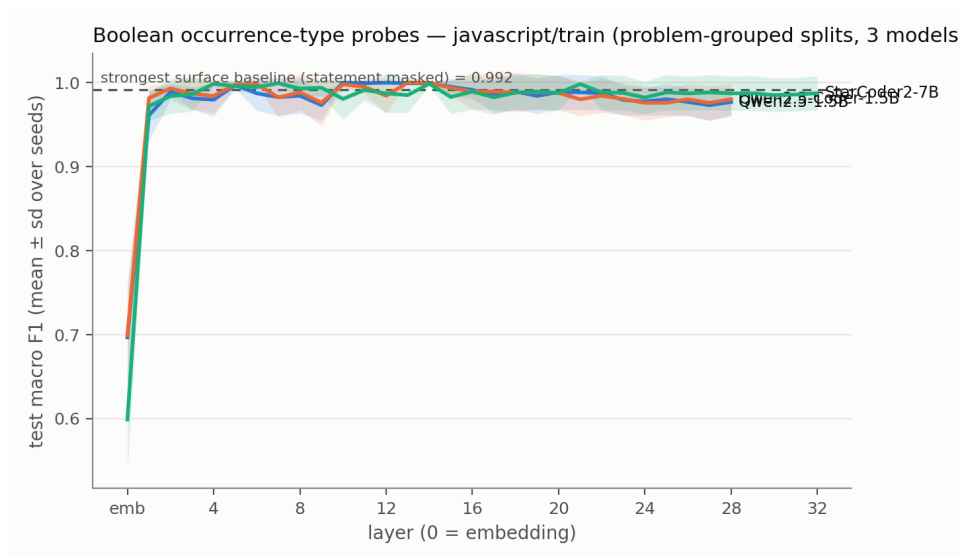


Figure 8: Boolean probe layer curves for JavaScript across the three models.

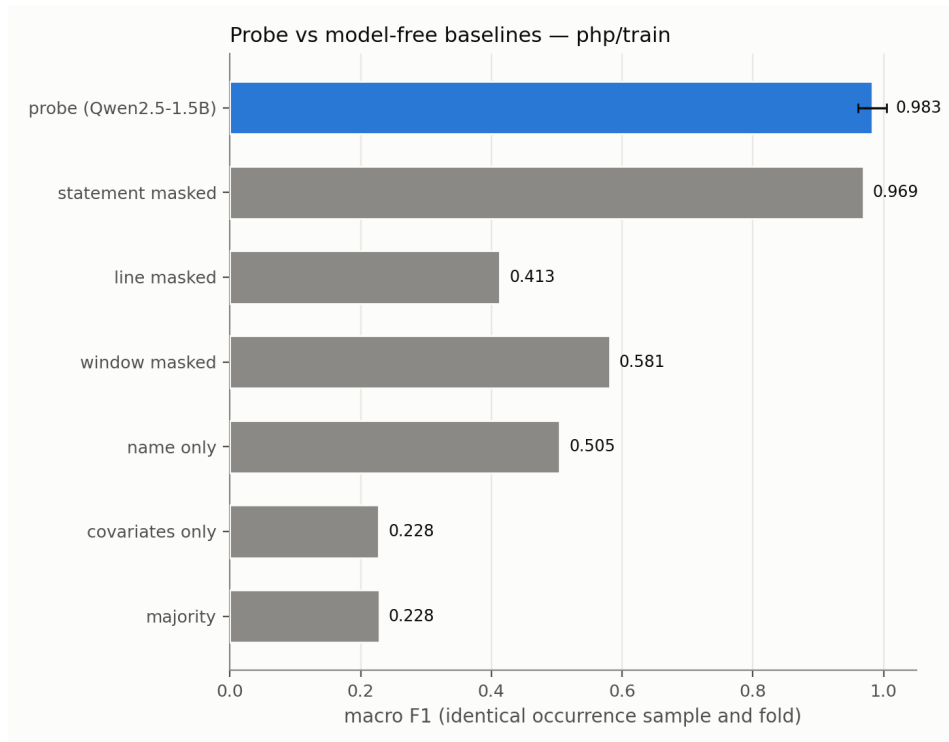


Figure 9: Boolean probe and model-free baselines for PHP, Qwen2.5-1.5B.

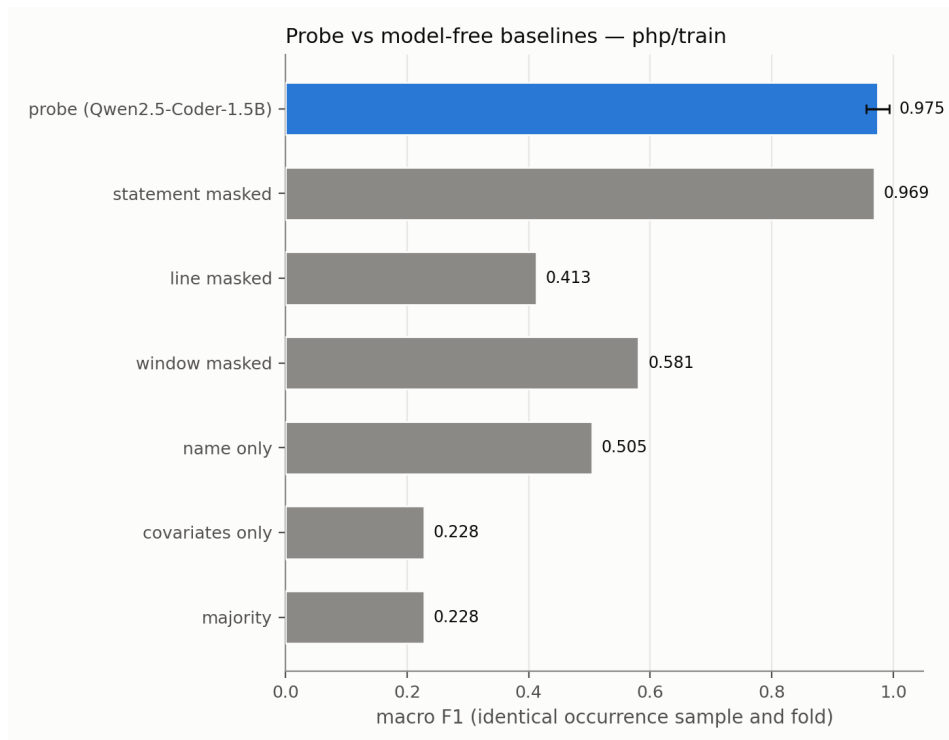


Figure 10: Boolean probe and model-free baselines for PHP, Qwen2.5-Coder-1.5B.

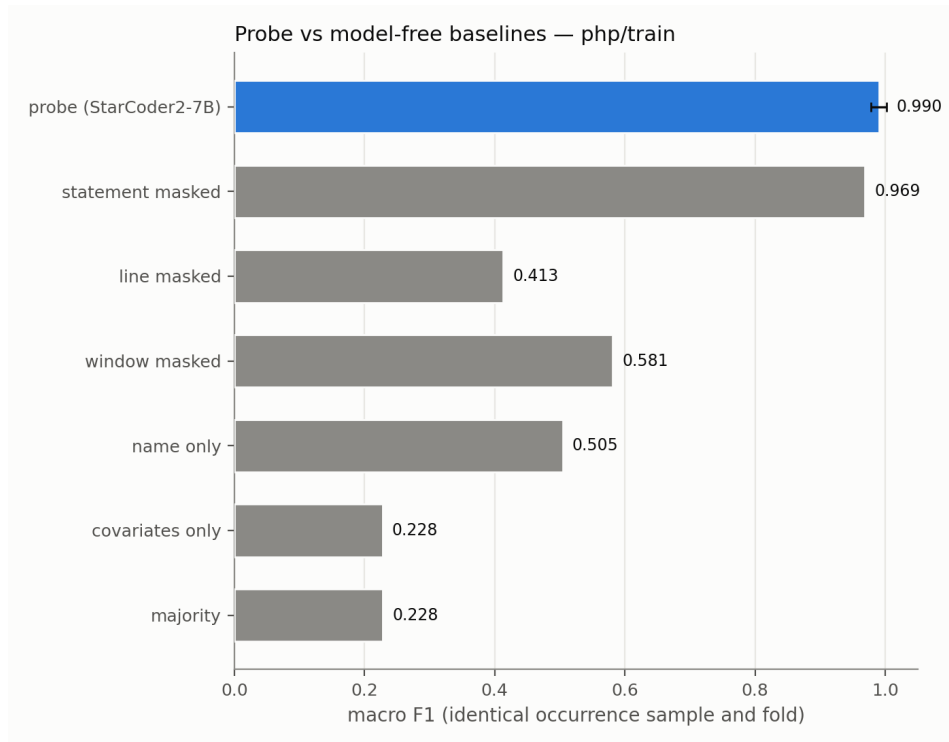


Figure 11: Boolean probe and model-free baselines for PHP, StarCoder2-7B.

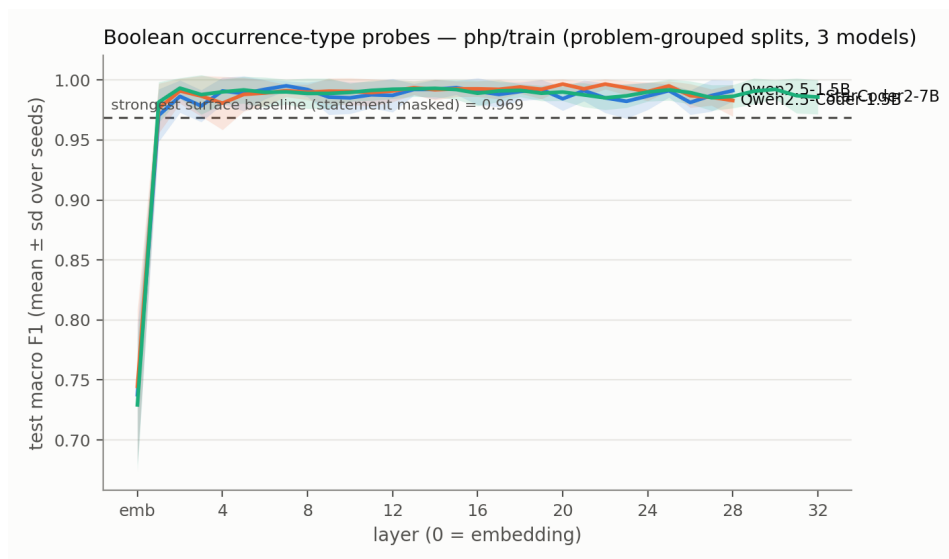


Figure 12: Boolean probe layer curves for PHP across the three models.

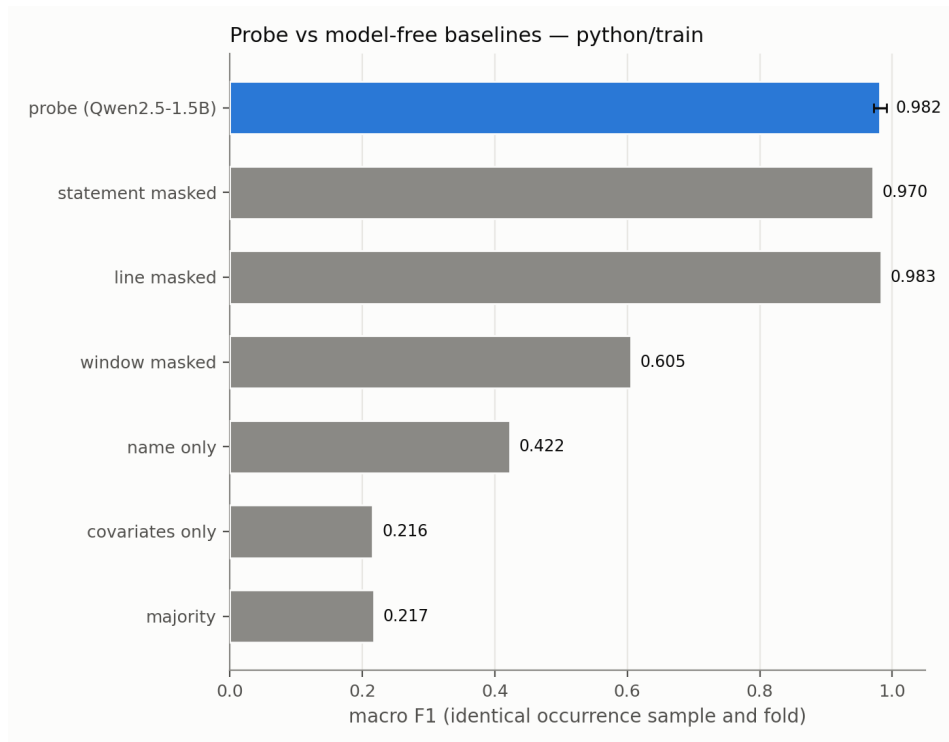


Figure 13: Boolean probe and model-free baselines for Python, Qwen2.5-1.5B.

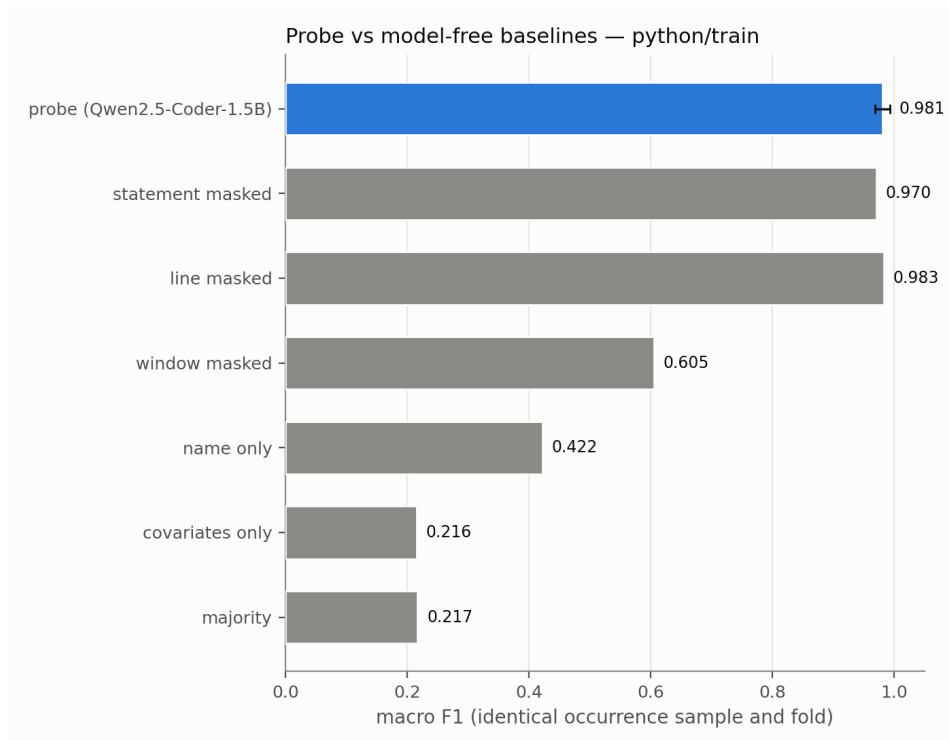


Figure 14: Boolean probe and model-free baselines for Python, Qwen2.5-Coder-1.5B.

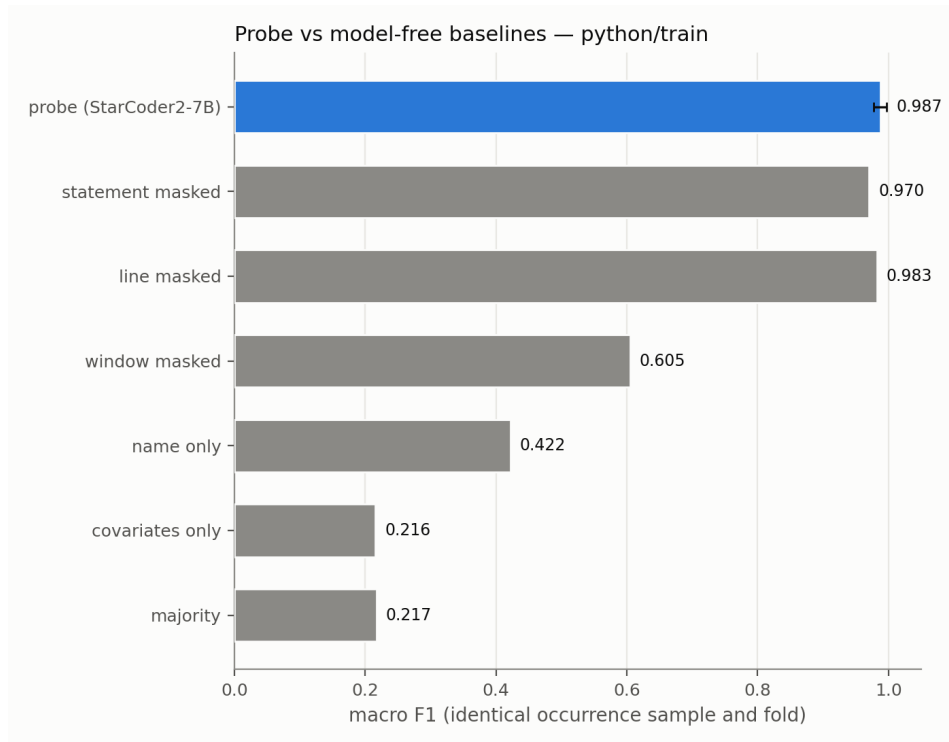


Figure 15: Boolean probe and model-free baselines for Python, StarCoder2-7B.

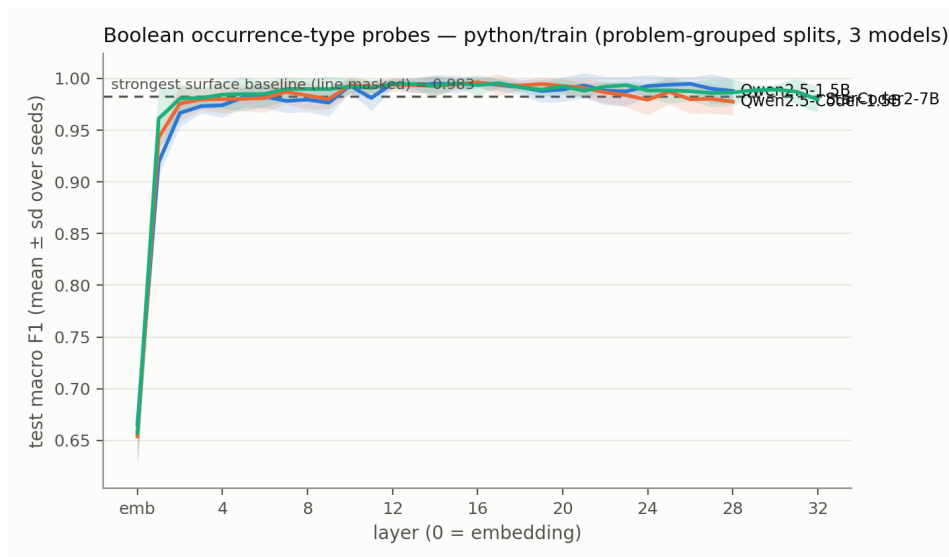


Figure 16: Boolean probe layer curves for Python across the three models.



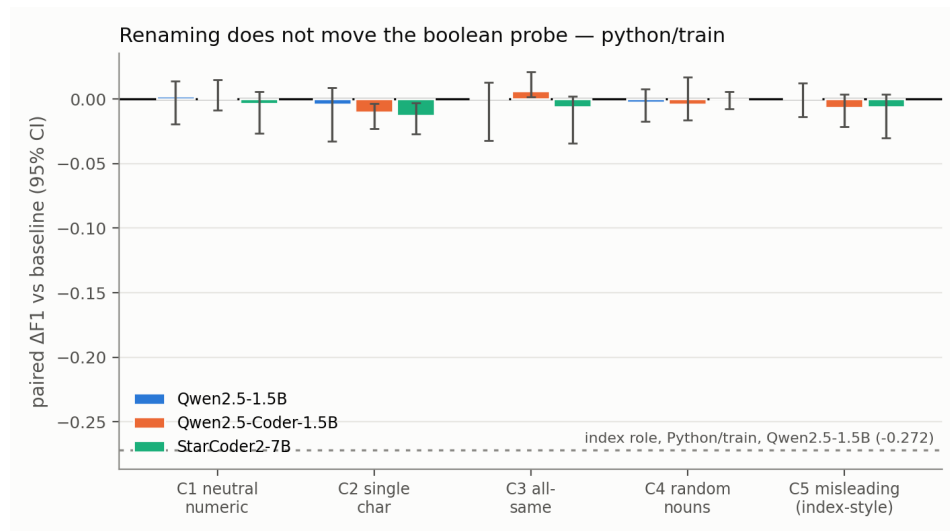


Figure 17: Problem-paired boolean refit-performance deltas on Python. No corresponding committed renaming run exists for other languages.

## 253 B Causal intervention details

254 **Status.** The iterator recovery exports predate the gate-specified class/structure protocol. They  
255 report descriptive recovery curves without clustered confidence intervals or matched placebo/random  
256 controls. The terminal value is one by construction, and C-language cross-language rows with zero  
257 pairs are missing data rather than null effects.

## 258 C Exploratory boolean interventions

259 **Status and estimand.** These boolean interventions are exploratory diagnostics rather than evidence  
260 for the main causal claim. The committed export contains layer-wise means but not the per-case  
261 arrays needed to recompute clustered intervals, and the runs do not use the specified gate of the  
262 class/structure experiment. The exporter defines effect as  $|\text{clean} - \text{intervened}|$  and specificity as that  
263 effect divided by the corresponding random-position-control effect. Specificity can become large  
264 when its denominator is small. We therefore report both specificity at the largest-effect layer and the  
265 maximum specificity over layers rather than selecting only the more favorable one.

266 **Omitted figures.** Per-cell diagnostic plots for the exploratory boolean interventions and the com-  
267 mitted role analyses are reproducible from the same artifacts with `make_interp_appendix.py`  
268 `-full-figures`. They visualize the tables above and add no uncertainty estimate the exports lack.

## 269 D Explicit coverage holes

- 270 • Accumulator and index/key results use legacy exports. No committed problem-grouped,  
271 five-seed CSV reproduces those roles under the modern protocol.
- 272 • Iterator results are available for Qwen2.5-Coder-1.5B and StarCoder2-7B, but not Qwen2.5-  
273 1.5B.
- 274 • Class/structure transfer exports include Python, C++, JavaScript, and C, but not Java, C#, or  
275 PHP.
- 276 • Boolean probe exports include Python, Java, JavaScript, and PHP. C, C#, and C++ are  
277 absent, and renaming is available only for Python.
- 278 • Gate-specified class/structure patching stopped after the Qwen2.5-1.5B null. Coder, Star-  
279 Coder2, and the all-layer core sweep were not attempted.
- 280 • No matched model-free surface comparator is committed for the class/structure role.

## 281 E Reproducibility

282 The appendix is generated from committed boolean, iterator, class or structure, and patching artifacts  
283 by `scripts/make_interp_appendix.py`. Modern probe conditions use frozen problem hashes,  
284 validation-selected layers, and five seeds where stated. Iterator summaries are single-run. The class  
285 or structure intervention records model and dataset revisions, prompt and configuration hashes, all  
286 4,032 primary rows, and clustered intervals. A regression test checks source traces for the headline  
287 numbers and rejects LP4FM-only conditions in this manuscript.

Table 8: Complete class/structure perturbation results under the shared five-seed protocol. Each condition refits a probe and selects its own layer on validation data, so cross-condition deltas do not test a fixed direction.

| model              | strategy                | layer | F1    | 95% CI         | control F1 | selectivity | $\Delta$ |
|--------------------|-------------------------|-------|-------|----------------|------------|-------------|----------|
| Qwen2.5-1.5B       | baseline                | L18   | 0.979 | [0.967, 0.987] | 0.427      | +0.552      | N/A      |
| Qwen2.5-1.5B       | random_nouns            | L20   | 0.986 | [0.976, 0.992] | 0.426      | +0.560      | +0.007   |
| Qwen2.5-1.5B       | single_chars            | L11   | 0.957 | [0.936, 0.969] | 0.423      | +0.534      | -0.022   |
| Qwen2.5-1.5B       | all_same                | L0    | 1.000 | [1.000, 1.000] | 0.440      | +0.560      | +0.021   |
| Qwen2.5-1.5B       | numeric_vars            | L21   | 0.965 | [0.952, 0.972] | 0.422      | +0.543      | -0.014   |
| Qwen2.5-1.5B       | misleading_class_struct | L7    | 0.976 | [0.962, 0.982] | 0.416      | +0.559      | -0.004   |
| Qwen2.5-Coder-1.5B | baseline                | L8    | 0.982 | [0.969, 0.989] | 0.429      | +0.553      | N/A      |
| Qwen2.5-Coder-1.5B | random_nouns            | L19   | 0.978 | [0.965, 0.985] | 0.427      | +0.551      | -0.004   |
| Qwen2.5-Coder-1.5B | single_chars            | L7    | 0.956 | [0.936, 0.967] | 0.423      | +0.533      | -0.026   |
| Qwen2.5-Coder-1.5B | all_same                | L0    | 1.000 | [1.000, 1.000] | 0.440      | +0.560      | +0.018   |
| Qwen2.5-Coder-1.5B | numeric_vars            | L21   | 0.968 | [0.955, 0.975] | 0.424      | +0.544      | -0.014   |
| Qwen2.5-Coder-1.5B | misleading_class_struct | L7    | 0.978 | [0.967, 0.984] | 0.415      | +0.562      | -0.004   |
| StarCoder2-7B      | baseline                | L5    | 0.981 | [0.967, 0.990] | 0.478      | +0.503      | N/A      |
| StarCoder2-7B      | random_nouns            | L6    | 0.976 | [0.962, 0.983] | 0.466      | +0.510      | -0.006   |
| StarCoder2-7B      | single_chars            | L4    | 0.957 | [0.921, 0.971] | 0.475      | +0.482      | -0.025   |
| StarCoder2-7B      | all_same                | L0    | 1.000 | [1.000, 1.000] | 0.445      | +0.555      | +0.019   |
| StarCoder2-7B      | numeric_vars            | L23   | 0.968 | [0.956, 0.975] | 0.451      | +0.517      | -0.013   |
| StarCoder2-7B      | misleading_class_struct | L5    | 0.985 | [0.976, 0.990] | 0.467      | +0.518      | +0.004   |

Table 9: Class/structure cross-language results. Transfer uses the Python-selected layer. In-domain layers are selected separately. Only languages present in the committed exports are shown.

| model              | language   | programs | in-domain layer | in-domain F1 | transfer accuracy | transfer F1 |
|--------------------|------------|----------|-----------------|--------------|-------------------|-------------|
| Qwen2.5-1.5B       | Python     | 400      | L18             | 0.979        | N/A               | N/A         |
| Qwen2.5-1.5B       | C++        | 606      | L10             | 0.991        | 0.993             | 0.933       |
| Qwen2.5-1.5B       | JavaScript | 285      | L5              | 0.987        | 0.998             | 0.957       |
| Qwen2.5-1.5B       | C          | 97       | L8              | 0.986        | 0.984             | 0.886       |
| Qwen2.5-Coder-1.5B | Python     | 400      | L8              | 0.982        | N/A               | N/A         |
| Qwen2.5-Coder-1.5B | C++        | 606      | L8              | 0.988        | 0.991             | 0.918       |
| Qwen2.5-Coder-1.5B | JavaScript | 285      | L6              | 0.985        | 0.996             | 0.913       |
| Qwen2.5-Coder-1.5B | C          | 97       | L20             | 0.980        | 0.983             | 0.885       |
| StarCoder2-7B      | Python     | 400      | L5              | 0.981        | N/A               | N/A         |
| StarCoder2-7B      | C++        | 606      | L5              | 0.992        | 0.996             | 0.965       |
| StarCoder2-7B      | JavaScript | 285      | L3              | 0.989        | 0.999             | 0.980       |
| StarCoder2-7B      | C          | 97       | L5              | 0.986        | 0.987             | 0.906       |

Table 10: Layerwise class/structure robustness summaries. The F1 range is taken over all committed layers. Maximum cosine similarity is measured against the baseline activation direction for each renamed condition.

| model              | strategy                | minimum F1 | maximum F1 (layer) | maximum cosine (layer) |
|--------------------|-------------------------|------------|--------------------|------------------------|
| Qwen2.5-1.5B       | all_same                | 0.999      | 1.000 (L0)         | 0.105 (L11)            |
| Qwen2.5-1.5B       | baseline                | 0.806      | 0.983 (L18)        | N/A                    |
| Qwen2.5-1.5B       | misleading_class_struct | 0.708      | 0.980 (L7)         | 0.311 (L15)            |
| Qwen2.5-1.5B       | numeric_vars            | 0.445      | 0.966 (L21)        | 0.329 (L22)            |
| Qwen2.5-1.5B       | random_nouns            | 0.529      | 0.989 (L15)        | 0.493 (L26)            |
| Qwen2.5-1.5B       | single_chars            | 0.530      | 0.965 (L7)         | 0.392 (L15)            |
| Qwen2.5-Coder-1.5B | all_same                | 0.999      | 1.000 (L0)         | 0.126 (L28)            |
| Qwen2.5-Coder-1.5B | baseline                | 0.807      | 0.985 (L7)         | N/A                    |
| Qwen2.5-Coder-1.5B | misleading_class_struct | 0.708      | 0.978 (L7)         | 0.324 (L7)             |
| Qwen2.5-Coder-1.5B | numeric_vars            | 0.445      | 0.969 (L24)        | 0.352 (L22)            |
| Qwen2.5-Coder-1.5B | random_nouns            | 0.530      | 0.983 (L23)        | 0.464 (L19)            |
| Qwen2.5-Coder-1.5B | single_chars            | 0.530      | 0.959 (L7)         | 0.415 (L15)            |
| StarCoder2-7B      | all_same                | 0.999      | 1.000 (L0)         | 0.101 (L21)            |
| StarCoder2-7B      | baseline                | 0.763      | 0.984 (L6)         | N/A                    |
| StarCoder2-7B      | misleading_class_struct | 0.717      | 0.985 (L3)         | 0.387 (L3)             |
| StarCoder2-7B      | numeric_vars            | 0.443      | 0.971 (L26)        | 0.369 (L22)            |
| StarCoder2-7B      | random_nouns            | 0.502      | 0.979 (L8)         | 0.511 (L3)             |
| StarCoder2-7B      | single_chars            | 0.529      | 0.962 (L3)         | 0.467 (L6)             |

Table 11: Complete iterator perturbation results. Each condition refits a probe and selects its own best layer. These single-run summaries do not carry the five-seed intervals used for the class/structure role.

| model              | strategy            | best layer | accuracy | F1    | $\Delta$ |
|--------------------|---------------------|------------|----------|-------|----------|
| Qwen2.5-Coder-1.5B | baseline            | L6         | 0.998    | 0.988 | N/A      |
| Qwen2.5-Coder-1.5B | random_nouns        | L9         | 0.996    | 0.974 | -0.014   |
| Qwen2.5-Coder-1.5B | single_chars        | L9         | 0.985    | 0.923 | -0.064   |
| Qwen2.5-Coder-1.5B | all_same            | L26        | 1.000    | 1.000 | +0.012   |
| Qwen2.5-Coder-1.5B | numeric_vars        | L23        | 0.978    | 0.924 | -0.064   |
| Qwen2.5-Coder-1.5B | misleading_iterator | L3         | 1.000    | 1.000 | +0.012   |
| StarCoder2-7B      | baseline            | L5         | 0.999    | 0.990 | N/A      |
| StarCoder2-7B      | random_nouns        | L7         | 0.997    | 0.986 | -0.005   |
| StarCoder2-7B      | single_chars        | L5         | 0.993    | 0.961 | -0.029   |
| StarCoder2-7B      | all_same            | L4         | 1.000    | 1.000 | +0.009   |
| StarCoder2-7B      | numeric_vars        | L24        | 0.989    | 0.958 | -0.032   |
| StarCoder2-7B      | misleading_iterator | L4         | 1.000    | 0.999 | +0.009   |

Table 12: Complete iterator cross-language transfer results. Transfer columns use the Python-selected layer.

| model              | language   | programs | in-domain layer | in-domain F1 | transfer accuracy | transfer F1 |
|--------------------|------------|----------|-----------------|--------------|-------------------|-------------|
| Qwen2.5-Coder-1.5B | Python     | 256      | L7              | 0.988        | N/A               | N/A         |
| Qwen2.5-Coder-1.5B | Java       | 188      | L12             | 0.996        | 0.991             | 0.950       |
| Qwen2.5-Coder-1.5B | C++        | 191      | L7              | 0.994        | 0.994             | 0.967       |
| Qwen2.5-Coder-1.5B | C          | 191      | L12             | 0.991        | 0.988             | 0.936       |
| Qwen2.5-Coder-1.5B | C#         | 189      | L13             | 0.998        | 0.992             | 0.957       |
| Qwen2.5-Coder-1.5B | JavaScript | 266      | L12             | 0.992        | 0.995             | 0.978       |
| Qwen2.5-Coder-1.5B | PHP        | 253      | L8              | 0.995        | 0.997             | 0.983       |
| StarCoder2-7B      | Python     | 256      | L10             | 0.991        | N/A               | N/A         |
| StarCoder2-7B      | Java       | 188      | L3              | 0.997        | 0.989             | 0.929       |
| StarCoder2-7B      | C++        | 191      | L5              | 0.997        | 0.991             | 0.941       |
| StarCoder2-7B      | C          | 191      | L16             | 0.992        | 0.984             | 0.906       |
| StarCoder2-7B      | C#         | 189      | L11             | 0.995        | 0.988             | 0.923       |
| StarCoder2-7B      | JavaScript | 266      | L16             | 0.998        | 0.986             | 0.931       |
| StarCoder2-7B      | PHP        | 253      | L17             | 0.998        | 0.993             | 0.964       |

Table 13: Layerwise iterator robustness summaries. F1 ranges cover every committed layer. Cosine similarity compares each perturbation direction with the baseline direction.

| model              | strategy            | minimum F1 | maximum F1 (layer) | maximum cosine (layer) |
|--------------------|---------------------|------------|--------------------|------------------------|
| Qwen2.5-Coder-1.5B | all_same            | 1.000      | 1.000 (L26)        | 0.334 (L0)             |
| Qwen2.5-Coder-1.5B | baseline            | 0.905      | 0.988 (L6)         | N/A                    |
| Qwen2.5-Coder-1.5B | misleading_iterator | 0.991      | 1.000 (L3)         | 0.187 (L15)            |
| Qwen2.5-Coder-1.5B | numeric_vars        | 0.556      | 0.924 (L23)        | 0.253 (L19)            |
| Qwen2.5-Coder-1.5B | random_nouns        | 0.633      | 0.974 (L9)         | 0.360 (L16)            |
| Qwen2.5-Coder-1.5B | single_chars        | 0.664      | 0.923 (L9)         | 0.498 (L0)             |
| StarCoder2-7B      | all_same            | 0.999      | 1.000 (L4)         | 0.337 (L9)             |
| StarCoder2-7B      | baseline            | 0.898      | 0.990 (L5)         | N/A                    |
| StarCoder2-7B      | misleading_iterator | 0.993      | 0.999 (L4)         | 0.265 (L6)             |
| StarCoder2-7B      | numeric_vars        | 0.552      | 0.958 (L24)        | 0.248 (L13)            |
| StarCoder2-7B      | random_nouns        | 0.613      | 0.986 (L7)         | 0.396 (L8)             |
| StarCoder2-7B      | single_chars        | 0.670      | 0.961 (L5)         | 0.451 (L0)             |

Table 14: Complete gate-specified class/structure patching summary in Qwen2.5-1.5B. The causal gate required mean effect at least 0.10, a positive interval, separation from controls, and recovery at least 0.05 in both directions.

| layer | span       | direction | control     | $n$ | mean effect | 95% CI             | recovery | minus random |
|-------|------------|-----------|-------------|-----|-------------|--------------------|----------|--------------|
| 0     | query_name | denoise   | target      | 288 | +0.0000     | [0.0000, 0.0000]   | +0.0000  | N/A          |
| 0     | query_name | noise     | target      | 288 | +0.0000     | [0.0000, 0.0000]   | +0.0000  | N/A          |
| 18    | placebo    | denoise   | target      | 288 | +0.0015     | [-0.0010, 0.0043]  | +0.0015  | N/A          |
| 18    | placebo    | noise     | target      | 288 | +0.0006     | [-0.0019, 0.0032]  | +0.0006  | N/A          |
| 18    | query_name | denoise   | random      | 288 | -0.0107     | [-0.0205, -0.0014] | -0.0110  | N/A          |
| 18    | query_name | denoise   | same_source | 288 | +0.0004     | [-0.0009, 0.0016]  | +0.0004  | N/A          |
| 18    | query_name | denoise   | target      | 288 | +0.0087     | [-0.0007, 0.0188]  | +0.0089  | +0.0194      |
| 18    | query_name | noise     | random      | 288 | +0.0096     | [0.0023, 0.0164]   | +0.0099  | N/A          |
| 18    | query_name | noise     | same_source | 288 | -0.0005     | [-0.0018, 0.0006]  | -0.0006  | N/A          |
| 18    | query_name | noise     | target      | 288 | +0.0199     | [0.0109, 0.0296]   | +0.0204  | +0.0103      |

Table 15: Aggregate probe-link diagnostic for the class/structure evaluation pairs. Absolute values are reported because pair orientation is arbitrary. Large probe movement alongside small behavioral movement is descriptive of the failed causal transfer, not evidence that probe movement causes behavior.

| quantity            | mean absolute | median absolute |
|---------------------|---------------|-----------------|
| probe movement      | 15.786        | 16.302          |
| behavioral movement | 0.038         | 0.031           |

Table 16: Class/structure gate and readout audit. Both prompt classes favor True. The pairwise gap remains separable, but the binary readout is not calibrated to distinguish function prompts.

| diagnostic                  | estimate    | 95% CI         | gate |
|-----------------------------|-------------|----------------|------|
| class logit difference      | 2.752       | [2.624, 2.898] | pass |
| function logit difference   | 1.778       | [1.652, 1.909] | fail |
| class-minus-function gap    | 0.974       | [0.908, 1.035] | pass |
| pair-gap accuracy           | 0.986       | [0.962, 0.993] | pass |
| query/declaration probe AUC | 0.943/0.905 | N/A            | pass |

Table 17: Patching audit trail. Completeness establishes that the scheduled Qwen evaluation finished. It does not turn the failed causal gate into positive evidence. The controller stopped before Coder and StarCoder2 after the Qwen null.

| audit item                           | value                |
|--------------------------------------|----------------------|
| behavior rows present/expected       | 1152/1152            |
| primary rows present/expected        | 4032/4032            |
| baseline probe-gap/behavior Spearman | 0.164 [0.023, 0.286] |
| controller terminal state            | stopped_qwen_null    |

Table 18: Exploratory iterator activation-patching recovery for Qwen2.5-Coder-1.5B, within-language perturbations.

| condition           | readout | pairs | clean/corrupt | L1    | L2    | L3    | L4    | L5    | L6    |
|---------------------|---------|-------|---------------|-------|-------|-------|-------|-------|-------|
| random_nouns        | L6      | 150   | 0.999/0.138   | 0.991 | 0.930 | 0.994 | 0.996 | 0.998 | 1.000 |
| single_chars        | L6      | 150   | 0.996/0.614   | 0.860 | 0.799 | 0.935 | 0.963 | 0.983 | 1.000 |
| all_same            | L6      | 150   | 0.999/0.142   | 0.808 | 0.900 | 0.961 | 0.986 | 0.998 | 1.000 |
| numeric_vars        | L6      | 150   | 0.999/0.057   | 0.966 | 0.961 | 0.991 | 0.997 | 0.999 | 1.000 |
| misleading_iterator | L6      | 150   | 0.999/0.286   | 0.820 | 0.835 | 0.916 | 0.946 | 0.983 | 1.000 |

Table 19: Exploratory iterator activation-patching recovery for Qwen2.5-Coder-1.5B, cross-language transfer.

| condition  | readout | pairs | clean/corrupt | L1     | L2     | L3    | L4    | L5    | L6    | L7    |
|------------|---------|-------|---------------|--------|--------|-------|-------|-------|-------|-------|
| Java       | L7      | 10    | 0.989/0.709   | -0.300 | -0.146 | 0.616 | 0.867 | 0.845 | 0.940 | 1.000 |
| C++        | L7      | 11    | 0.979/0.769   | -0.066 | 0.025  | 0.785 | 0.869 | 0.869 | 0.904 | 1.000 |
| C          | L7      | 0     | 0.000/0.000   | N/A    | N/A    | N/A   | N/A   | N/A   | N/A   | N/A   |
| C#         | L7      | 9     | 0.981/0.629   | 0.307  | 0.353  | 0.772 | 0.867 | 0.903 | 0.989 | 1.000 |
| JavaScript | L7      | 13    | 0.983/0.766   | 0.708  | 0.678  | 0.968 | 0.943 | 0.953 | 0.983 | 1.000 |
| PHP        | L7      | 7     | 0.982/0.748   | -0.211 | -0.397 | 0.614 | 0.760 | 0.934 | 1.027 | 1.000 |

Table 20: Exploratory iterator activation-patching recovery for StarCoder2-7B, within-language perturbations.

| condition           | readout | pairs | clean/corrupt | L1    | L2    | L3    | L4    | L5    |
|---------------------|---------|-------|---------------|-------|-------|-------|-------|-------|
| random_nouns        | L5      | 150   | 0.998/0.569   | 0.931 | 0.766 | 0.966 | 0.951 | 1.000 |
| single_chars        | L5      | 150   | 0.994/0.686   | 0.636 | 0.006 | 0.823 | 0.759 | 1.000 |
| all_same            | L5      | 150   | 0.999/0.249   | 0.342 | 0.772 | 0.946 | 0.979 | 1.000 |
| numeric_vars        | L5      | 150   | 0.998/0.530   | 0.889 | 0.811 | 0.953 | 0.955 | 1.000 |
| misleading_iterator | L5      | 150   | 0.997/0.628   | 0.777 | 0.104 | 0.921 | 0.843 | 1.000 |

Table 21: Exploratory iterator activation-patching recovery for StarCoder2-7B, cross-language transfer.

| condition  | readout | pairs | clean/corrupt | L1    | L2    | L3    | L4    | L5    | L6    | L7    | L8    | L9    | L10   |
|------------|---------|-------|---------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Java       | L10     | 150   | 0.996/0.750   | 0.254 | 0.634 | 0.829 | 0.913 | 0.957 | 0.964 | 0.979 | 0.988 | 0.994 | 1.000 |
| C++        | L10     | 150   | 0.996/0.771   | 0.337 | 0.756 | 0.892 | 0.938 | 0.966 | 0.969 | 0.979 | 0.988 | 0.995 | 1.000 |
| C          | L10     | 0     | 0.000/0.000   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
| C#         | L10     | 150   | 0.996/0.740   | 0.294 | 0.692 | 0.877 | 0.942 | 0.969 | 0.974 | 0.984 | 0.991 | 0.996 | 1.000 |
| JavaScript | L10     | 150   | 0.996/0.716   | 0.311 | 0.743 | 0.880 | 0.939 | 0.972 | 0.976 | 0.983 | 0.990 | 0.996 | 1.000 |
| PHP        | L10     | 73    | 0.993/0.808   | 0.491 | 0.796 | 0.938 | 0.964 | 0.979 | 0.973 | 0.983 | 0.994 | 0.999 | 1.000 |

Table 22: Exploratory boolean patch diagnostics. Effect and specificity use the exporter definitions above. No uncertainty interval is available from the committed summary.

| language   | model              | $n$ | peak layer | peak effect | spec. at peak | best spec. (layer) |
|------------|--------------------|-----|------------|-------------|---------------|--------------------|
| C++        | Qwen2.5-1.5B       | 157 | L0         | 14.845      | $3.6\times$   | $7.9\times$ (L20)  |
| C++        | Qwen2.5-Coder-1.5B | 158 | L0         | 15.414      | $3.6\times$   | $6.9\times$ (L8)   |
| C++        | StarCoder2-7B      | 157 | L4         | 14.982      | $8.1\times$   | $10.4\times$ (L16) |
| Java       | Qwen2.5-1.5B       | 155 | L8         | 11.607      | $6.1\times$   | $7.7\times$ (L12)  |
| Java       | Qwen2.5-Coder-1.5B | 155 | L8         | 12.642      | $5.9\times$   | $7.7\times$ (L12)  |
| Java       | StarCoder2-7B      | 155 | L4         | 13.900      | $7.5\times$   | $8.6\times$ (L12)  |
| JavaScript | Qwen2.5-1.5B       | 58  | L8         | 12.755      | $6.9\times$   | $6.9\times$ (L8)   |
| JavaScript | Qwen2.5-Coder-1.5B | 62  | L8         | 12.717      | $6.6\times$   | $7.4\times$ (L12)  |
| JavaScript | StarCoder2-7B      | 129 | L4         | 12.137      | $8.1\times$   | $8.2\times$ (L16)  |
| PHP        | Qwen2.5-1.5B       | 15  | L4         | 11.698      | $7.2\times$   | $7.2\times$ (L4)   |
| PHP        | Qwen2.5-Coder-1.5B | 16  | L4         | 11.425      | $4.7\times$   | $7.4\times$ (L0)   |
| PHP        | StarCoder2-7B      | 17  | L4         | 11.545      | $6.3\times$   | $9.3\times$ (L16)  |
| Python     | Qwen2.5-1.5B       | 176 | L8         | 12.948      | $6.0\times$   | $7.2\times$ (L16)  |
| Python     | Qwen2.5-Coder-1.5B | 176 | L8         | 13.275      | $6.7\times$   | $7.8\times$ (L16)  |
| Python     | StarCoder2-7B      | 176 | L4         | 13.592      | $7.9\times$   | $9.7\times$ (L12)  |

Table 23: Exploratory boolean steer diagnostics. Effect and specificity use the exporter definitions above. No uncertainty interval is available from the committed summary.

| language   | model              | $n$ | peak layer | peak effect | spec. at peak | best spec. (layer) |
|------------|--------------------|-----|------------|-------------|---------------|--------------------|
| C++        | Qwen2.5-1.5B       | 79  | L24        | 2.508       | 45.0×         | 169.3× (L4)        |
| C++        | Qwen2.5-Coder-1.5B | 79  | L24        | 1.436       | 602.0×        | 602.0× (L24)       |
| C++        | StarCoder2-7B      | 78  | L20        | 0.653       | 6.9×          | 345.4× (L4)        |
| Java       | Qwen2.5-1.5B       | 78  | L24        | 2.486       | 72.2×         | 826.9× (L8)        |
| Java       | Qwen2.5-Coder-1.5B | 78  | L24        | 1.712       | 72.0×         | 179.9× (L0)        |
| Java       | StarCoder2-7B      | 78  | L24        | 0.555       | 244.0×        | 244.0× (L24)       |
| JavaScript | Qwen2.5-1.5B       | 31  | L24        | 1.763       | 24.5×         | 76.8× (L8)         |
| JavaScript | Qwen2.5-Coder-1.5B | 31  | L24        | 1.147       | 65.3×         | 147.6× (L4)        |
| JavaScript | StarCoder2-7B      | 65  | L16        | 0.591       | 11.2×         | 61.0× (L28)        |
| PHP        | Qwen2.5-1.5B       | 7   | L24        | 2.759       | 190.2×        | 190.2× (L24)       |
| PHP        | Qwen2.5-Coder-1.5B | 8   | L24        | 0.898       | 11.5×         | 132.7× (L8)        |
| PHP        | StarCoder2-7B      | 9   | L20        | 0.991       | 8.7×          | 99.9× (L4)         |
| Python     | Qwen2.5-1.5B       | 88  | L24        | 2.175       | 209.1×        | 794.6× (L16)       |
| Python     | Qwen2.5-Coder-1.5B | 88  | L24        | 1.231       | 1027.1×       | 1027.1× (L24)      |
| Python     | StarCoder2-7B      | 88  | L20        | 0.721       | 8.6×          | 55.1× (L4)         |

Table 24: Exploratory boolean ablate diagnostics. Effect and specificity use the exporter definitions above. No uncertainty interval is available from the committed summary.

| language   | model              | $n$ | peak layer | peak effect | spec. at peak | best spec. (layer) |
|------------|--------------------|-----|------------|-------------|---------------|--------------------|
| C++        | Qwen2.5-1.5B       | 157 | L0         | 14.286      | 4.5×          | 31.9× (L4)         |
| C++        | Qwen2.5-Coder-1.5B | 158 | L0         | 12.927      | 4.4×          | 76.2× (L20)        |
| C++        | StarCoder2-7B      | 157 | L0         | 13.405      | 4.3×          | 36.5× (L4)         |
| Java       | Qwen2.5-1.5B       | 155 | L16        | 9.720       | 28.5×         | 86.0× (L12)        |
| Java       | Qwen2.5-Coder-1.5B | 155 | L4         | 9.331       | 12.3×         | 62.4× (L12)        |
| Java       | StarCoder2-7B      | 155 | L4         | 11.190      | 27.0×         | 27.0× (L4)         |
| JavaScript | Qwen2.5-1.5B       | 58  | L16        | 10.852      | 26.7×         | 26.7× (L16)        |
| JavaScript | Qwen2.5-Coder-1.5B | 62  | L16        | 9.868       | 28.1×         | 116.4× (L24)       |
| JavaScript | StarCoder2-7B      | 129 | L4         | 10.970      | 27.3×         | 27.3× (L4)         |
| PHP        | Qwen2.5-1.5B       | 15  | L16        | 15.698      | 15.0×         | 102.4× (L20)       |
| PHP        | Qwen2.5-Coder-1.5B | 16  | L16        | 13.402      | 26.5×         | 79.2× (L20)        |
| PHP        | StarCoder2-7B      | 17  | L4         | 14.260      | 19.4×         | 19.7× (L8)         |
| Python     | Qwen2.5-1.5B       | 176 | L0         | 12.144      | 5.8×          | 36.7× (L24)        |
| Python     | Qwen2.5-Coder-1.5B | 176 | L0         | 11.286      | 6.8×          | 33.1× (L16)        |
| Python     | StarCoder2-7B      | 176 | L4         | 11.818      | 26.5×         | 26.5× (L4)         |

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Justification: Large language models are the object of study, not an important or non-standard component used to develop the research method. Writing and editing assistance is exempt under the question.

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